Angel Martinez

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EDUCATION

New Mexico State University Las Cruces, NM

Bachelor of Science, Computer Science, Software Development Concentration. GPA: 3.29. *December 2026*

Relevant Coursework: Database Management Systems, Software Development, Modern Web Development, Computer Security

SKILLS

Technical Skills: Java (Advanced), Python (Advanced), C++ (Intermediate), C (Intermediate), MySQL (Intermediate)

Languages: English (Native), Spanish (Fluent)

EXPERIENCE

Research Assistant May 2026 – Present

New Mexico State University, Department of Astronomy *Las Cruces, NM*

- Apply Python to process and analyze astronomical data under a faculty principal investigator in support of ongoing research projects
- Execute data-driven research tasks assigned by the principal investigator, contributing directly to the department's active research efforts

Research Intern June 2026 – August 2026

NASA Goddard Space Flight Center (GSFC), OSTEM MOSAICS Program *Greenbelt, MD*

- Developed a Python-based shadow metrology pipeline to recover object height and 3D terrain slope from overhead imagery, implementing a solver adapted from lunar surface physics models
- Built reproducible workflows with NumPy, SciPy, and Astropy; validated pipeline against controlled data and delivered technical presentations for NASA principal investigators

PROJECTS

StudyAI April 2026

Hackathon Project

- Built a full-stack web app with Next.js and React that converts pasted study notes into AI-generated study material
- Engineered a Node.js API route integrating Groq's LLaMA 3.3 70B model to generate a concise summary and 12 question-and-answer flashcards per submission
- Added document-parsing support for PDF and Office file uploads using the pdf-parse and office-text-extractor libraries

Nuke-Map April 2026

Group Project, Modern Web Technologies Course

- Collaborated with a 5-person team to build a full-stack Node.js/Express web app visualizing the effects of a nuclear detonation at any location worldwide using the Leaflet.js mapping library
- Implemented the core blast-radius calculation engine, modeling fireball, blast-overpressure, thermal-radiation, and initial-radiation zones with yield-scaling formulas and rendering them as interactive map overlays
- Designed RESTful API endpoints backed by a MySQL database to persist and retrieve simulation data, including location, coordinates, and warhead yield

CERTIFICATIONS

- Certificate of Participation, Hackathon (2026)